Mohammad Shaheer

mshaheer@alumni.cmu.edu, mshaheer@andrew.cmu.edu | +1(412)287-1845, +97450507802

Education

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Computer Science - Research Thesis — May 2027

Carnegie Mellon University — **Doha, Qatar**
Bachelor of Science in Computer Science | Concentration in Algorithms & Complexity — May 2023
University Honors & College Honors
Relevant Coursework: Design and Analysis of Algorithms, Introduction to Computer Systems, Distributed Systems,

Skills

Programming Languages: Python, C/C++/C#, Java, SML, Typescript/Javascript
Frameworks & Databases: Django REST Framework/Django, .NET, Flutter, NextJS, MongoDB, PostgreSQL, React
Cloud & Tools: GCP, AWS, Azure, Heroku, Docker, Kubernetes, Linux

Professional Experience

Avey | Full Stack Software Engineer | Doha, Qatar — Dec. 2023 - May. 2026

- Developed Coder, an AI-powered insurance claim auditing platform to **automate error detection and reduce claim rejections**; deployed in 2 local hospitals supporting 4 insurance companies
- Worked directly with doctors, hospital staff, and insurance company representatives to translate clinical workflow requirements into system architecture and adjudication rules
- Engineered a token escrow system to prevent overcharging; **reducing monthly billing errors from ~30% to ~3%**
- Implemented async Celery workers with multi-priority queues for claim processing; **reducing end-to-end processing time from ~15 seconds to ~6 seconds** per claim
- Integrated cloud-deployed AI models into the medical adjudication pipeline, **enabling automated, real-time evaluation of claims and reducing reliance on manual review**
- Led the development of an internal HR portal **used by 50+ employees**, delivering a full-stack system (from architecture to deployment) in under 4 months
- Replaced ad hoc Slack-based approval processes with a centralized, audited system, **cutting average approval time from days to hours** across ~100+ monthly requests
- Optimized Django ORM queries on a high-traffic list endpoint, **reducing response time from ~900ms to ~45ms** by profiling query execution and eliminating N+1 patterns

Snoonu | Software Engineer | Doha, Qatar — May 2023 - Dec 2023

- Conducted EDA on marketing campaign data to assess feature suitability for demand prediction models
- **Improved testing accuracy of demand forecasting model by ~17%** through hyperparameter tuning
- Deployed a real-time driver tracking system communicating driver status updates to the merchant portal periodically via SignalR and RabbitMQ **reducing merchant calls to operations team by 37%**

Projects

Distributed File System | Java — Carnegie Mellon University | Nov. 2022

- Built a distributed file system using custom RMI with multi-threaded Skeleton/Stub architecture, **enabling remote file operations across multiple storage servers**
- Implemented a metadata management system on a centralized naming server, coordinating storage server registration, duplicate pruning, and synchronized create/delete operations across nodes
- Implemented hierarchical read-write locking with FIFO-ordered queues and root-to-leaf path locking, **ensuring consistency for concurrent multi-client reads and writes while preventing deadlocks**
- Engineered a dynamic replication engine that auto-scaled file replicas based on request load using a linear coarse-grained policy, with invalidation-based write propagation **to keep replicas consistent**

Dynamic Storage Allocator | C — Carnegie Mellon University | Nov. 2020

- Engineered a dynamic storage allocator using segregated free lists with next-fit placement, **achieving 20,634 KOPS throughput (34× libc baseline) at 67.6% space utilization**
- Implemented a heap consistency checker **to catch pointer corruption, coalescing failures, and invalid free-list invariants** during development